**NAME:** ____________________ **PERIOD:** ________ **DATE:** __________

# What Did Each Simulation Estimate?

AP Statistics · Unit 2 · Topic 2.3 · B: 2026-10-23 · E: 2026-10-23

## [STAR] TODAY'S PROBLEM

Suppose each of five students can be treated as independently selected, and each has probability 0.12 of being left-handed. Two students use 1,000 simulation trials to estimate the probability that at least 2 of the 5 students are left-handed. Their plans and results are shown.

**Results: Student 1 = 87/1,000; Student 2 = 94/1,000**

| Plan | $L$ assignment | Trial; success |
|---|---|---|
| **1** | 0 of 0–9 | 5 digits; 2+ L |
| **2** | 00–11 of 00–99 | 5 pairs; exactly 2 L |

### FIRST TAKE — before we discuss (5 min)

Which simulation result would you trust more for the question asked—Student 1, Student 2, or neither? Name one reason.

### [BOOK] THREE MATCHES MAKE A VALID SIMULATION (keep on the page)

Match each outcome's chance with the fraction of equally likely labels assigned to it. One trial must reproduce the stated number of independent selections. Count the exact event named in the question. More trials reduce chance variation but do not repair a wrong assignment or event. Estimate probability with successful trials/total trials.

**Work (a)–(c) from the rules box:**

**DOK 1** **(a)** List the left-handed counts included in at least 2 of 5. State the chance one student is left-handed.

**DOK 2** **(b)** Find each reported estimate. For each plan, identify either a mismatch in the chance assignment or a mismatch in the event counted.

**DOK 3** **(c)** Would you trust either plan for the question asked? Explain both mismatches and why the larger reported estimate does not validate a plan. Describe one corrected trial, including the random labels and the event counted. Write 3–4 sentences.

**Frame:** Plan 1 uses a chance of ________ instead of ________; Plan 2 leaves out ____________________.

**Frame:** My corrected trial uses ____________________________ and counts a success when ____________________________.

**EXIT — turn this in**

One thing the rules box changed about my first take: ______________________________________________